University of Moratuwa
Department of Computer Science and Engineering

AutoVaultLK

User Manual

Dealers

Second-Hand Vehicle Marketplace with Intelligent Search and Automated Inventory Processing

Table of Contents

1. Introduction 3
   1.1 Purpose and Scope 3
   1.2 System Overview 3
   1.3 Prerequisites and Conventions 3
2. Registration and Verification 3
   2.1 Dealer Types 3
   2.2 Registration 4
   2.3 Email Confirmation and Verification 5
3. Dealer Portal 5
   3.1 Navigation 5
   3.2 Dashboard 5
   3.3 Business Details 6
4. Managing Listings 6
   4.1 Listing Statuses 6
   4.2 Reviewing and Approving Imported Listings 7
   4.3 Editing, Archiving and Deleting 7
5. Manual Listing 8
   5.1 Listing Form 8
   5.2 Type-Specific Details, Description and Photographs 8
   5.3 Saving 9
6. Bulk Upload 9
   6.1 Overview 9
   6.2 Preparing the Inventory File 9
   6.3 Preparing the Photograph Archive 10
   6.4 Uploading 10
   6.5 Monitoring and Results 11
7. Troubleshooting 13

1. Introduction

1.1 Purpose and Scope

This manual describes the use of AutoVaultLK by dealers, that is, by the individuals and businesses that sell vehicles through the platform. It explains how a dealership is registered and verified, how listings are created, reviewed and maintained, and how an entire inventory is imported in one operation.The functions used by buyers and administrators are described in separate manuals.

1.2 System Overview

AutoVaultLK is a second-hand vehicle marketplace for Sri Lanka, developed as Project 11 by Group 25 of the Department of Computer Science and Engineering, University of Moratuwa. Its functional requirements are recorded in the Software Requirements Specification [2], and its design is described in the Software Architecture Document [3]. Dealers work in a dedicated Dealer Portal, which is reached from the **Dealers** item of the main menu of the public site. The portal allows vehicles to be listed one at a time or imported in bulk, and it applies automatic cleaning to imported data so that misspelled makes and models are corrected before the vehicles are published.

1.3 Prerequisites and Conventions

A current web browser and an email address that can be opened are required. A business dealer additionally requires a business registration certificate in PDF, JPEG or PNG format of at most 5 MB, while an individual dealer requires a National Identity Card (NIC) number. Bulk upload, which is available to business dealers only, requires an inventory file and, optionally, an archive of photographs, as described in Section 6. The names of buttons, tabs and fields are printed in bold as they appear on screen, and all data shown in the screen captures are sample data.

2. Registration and Verification

2.1 Dealer Types

A dealer account is either individual or business, and the type is chosen at registration and cannot be changed afterwards. The differences are summarised in Table 1. An individual dealer is a person selling in a personal capacity and is identified by an NIC number, whereas a business dealer is a registered dealership that is identified by a registration number and a certificate. Only business dealers may use bulk upload and the upload history.

Table 1. Dealer types and their requirements.

Item	Business dealer	Individual dealer
Identification	Registration number and uploaded certificate (PDF, JPEG or PNG, up to 5 MB).	NIC number: nine digits followed by V or X, or twelve digits.
Registration number	Mandatory.	Optional.

Item	Business dealer	Individual dealer
Portal functions	Dashboard, listings, manual listing, bulk upload, upload history and business details.	Dashboard, listings, manual listing and business details.

2.2 Registration

Registration is started with **Register as dealer** on the home page or with the registration link on the Dealer Portal sign-in page, shown in Fig. 1. The registration is a four-step wizard with a progress bar, and each step is validated before the wizard proceeds. The first step collects the company name, the dealer type, the registration number, the address and the city, together with the certificate for a business or the NIC number for an individual. The second step records the contact name and a telephone number, which is entered without the leading zero beside a country code. The third step sets the email address and password, which must contain at least eight characters with a lowercase letter, an uppercase letter and a digit. The fourth step presents a summary, shown on the right of Fig. 2, and the registration is submitted with **Create Dealer Account**.

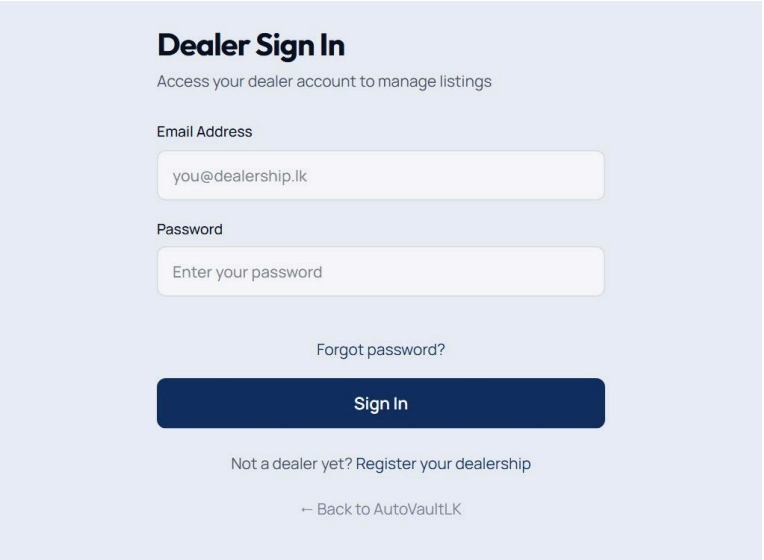


Fig. 1. Dealer Portal sign-in page.

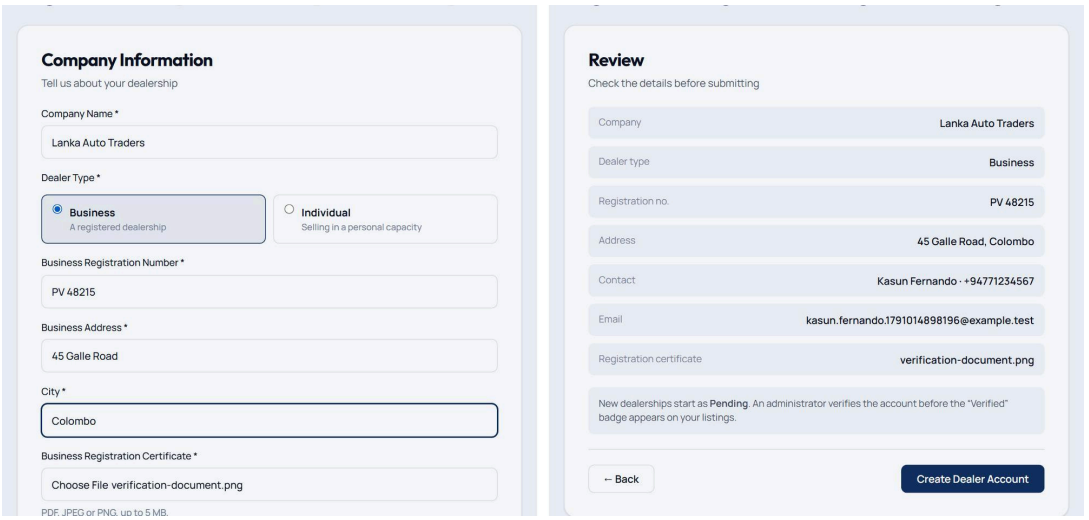


Fig. 2. Dealer registration: company information (left) and review (right).

2.3 Email Confirmation and Verification

After submission, a confirmation notice appears and a verification email is sent. Once the link in that email has been opened, the dealer can sign in at once. The application, however, remains pending until an administrator has reviewed it, and the sequence of states is shown in Fig. 3. An application is approved, which verifies the dealer, or rejected with a reason. A rejected dealer may correct the details and resubmit, which returns the application to the pending state.

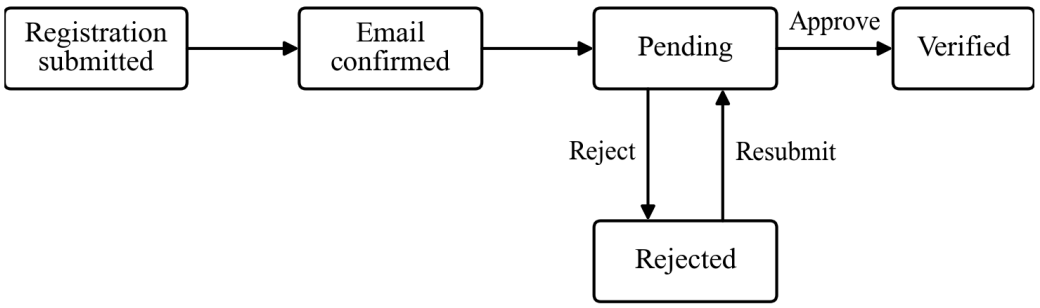


Fig. 3. Verification states of a dealer account.

While the application is pending, the portal shows only the notice in Fig. 4, and no menu is available. After approval, the full portal opens and a **Verified** badge appears on the dealer's listings throughout the platform. After a rejection, the notice displays the reason given by the administrator together with a form for correcting the company details and uploading the certificate or entering the NIC again. The **Resubmit for review** button sends the application back for review.

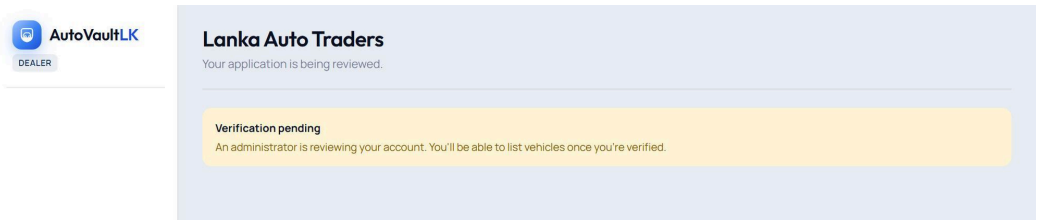


Fig. 4. Notice displayed while an application is pending.

3. Dealer Portal

3.1 Navigation

The sidebar of the portal offers **Dashboard**, **My listings**, **Manual listing** and **Business details** and, for business dealers only, **Bulk upload** and **Upload history**. The signed-in email address and a **Sign out** button are placed at the foot of the sidebar. Signing out should always be carried out on a shared computer.

3.2 Dashboard

The dashboard, shown in Fig. 5, summarises the stock of the dealer. A row of tiles gives the numbers of live, pending review, draft, rejected and archived listings and the live inventory value, which is the total asking price of the live listings in Sri Lankan rupees (LKR). The five most recent listings are shown with their status, and a chart displays the distribution of the inventory by make. The **My listings** link opens the complete list.

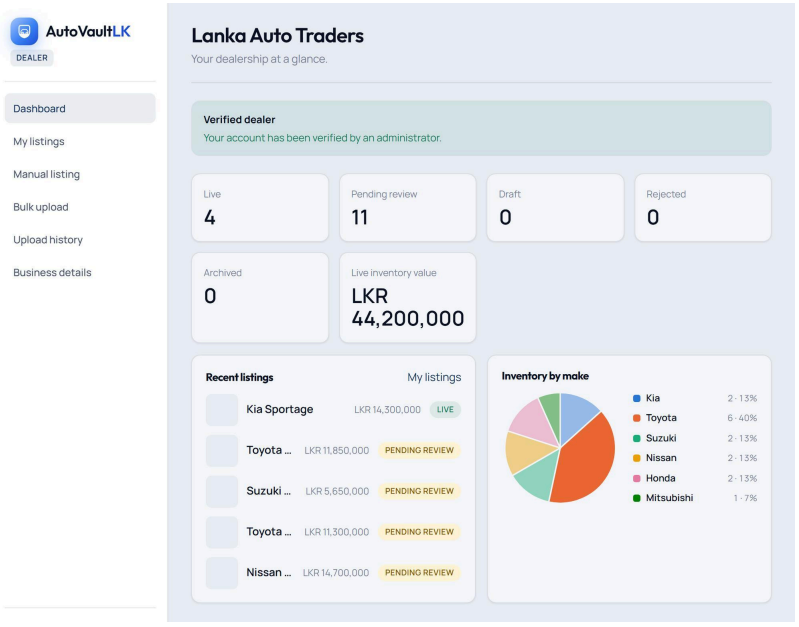


Fig. 5. Dealer dashboard.

3.3 Business Details

The **Business details** page allows the contact name, company name, registration number (business dealers only), address, city and contact number to be edited. The contact number is entered in international form, for example +94771234567. The **Save changes** button becomes

active after a field has been changed, and a message confirms the update. The email address and the dealer type cannot be altered after registration.

4. Managing Listings

4.1 Listing Statuses

The **My listings** page lists every vehicle owned by the dealer, however it was added, with its type, condition, year, price, mileage and status. A listing passes through the statuses shown in Fig. 6, in which each arrow is an action available to the dealer. A live listing is visible to buyers, whereas pending review, draft, rejected and archived listings are not. Vehicles imported by bulk upload begin as pending review so that they can be checked before publication. Manual listings created by a verified dealer are published immediately. A vehicle is recorded as sold by archiving it, which removes it from search, and an archived listing can be restored. A listing that has never been published can be deleted permanently, but a published listing can only be archived, because buyers may already have saved it.

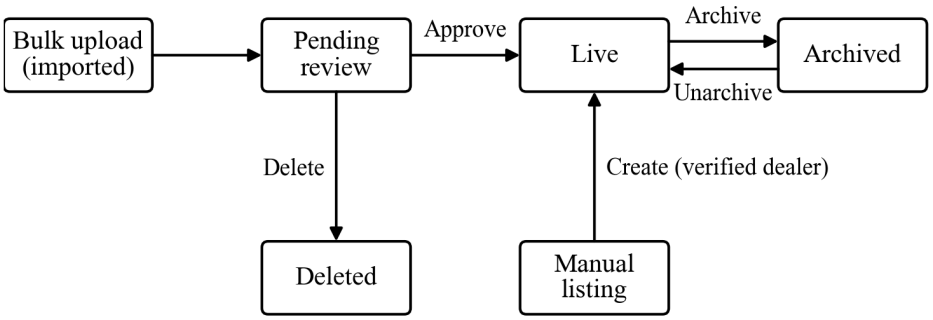


Fig. 6. Life cycle of a listing.

4.2 Reviewing and Approving Imported Listings

A banner at the top of the page states how many listings await review, as shown in Fig. 7. The default sort order places the rows that the cleaning process was least confident about first, and the order can be changed with the **Sort by** menu or by selecting a column heading. Beneath each row, **Show field details** explains how every imported value was interpreted and labels it as parsed, matched or corrected automatically, together with a plain-language note such as a corrected make or an inferred fuel type. **Show specs & photos** displays the stored specifications and the photographs matched to the vehicle, and a vehicle that has no registration number is flagged because photographs are matched on that number.

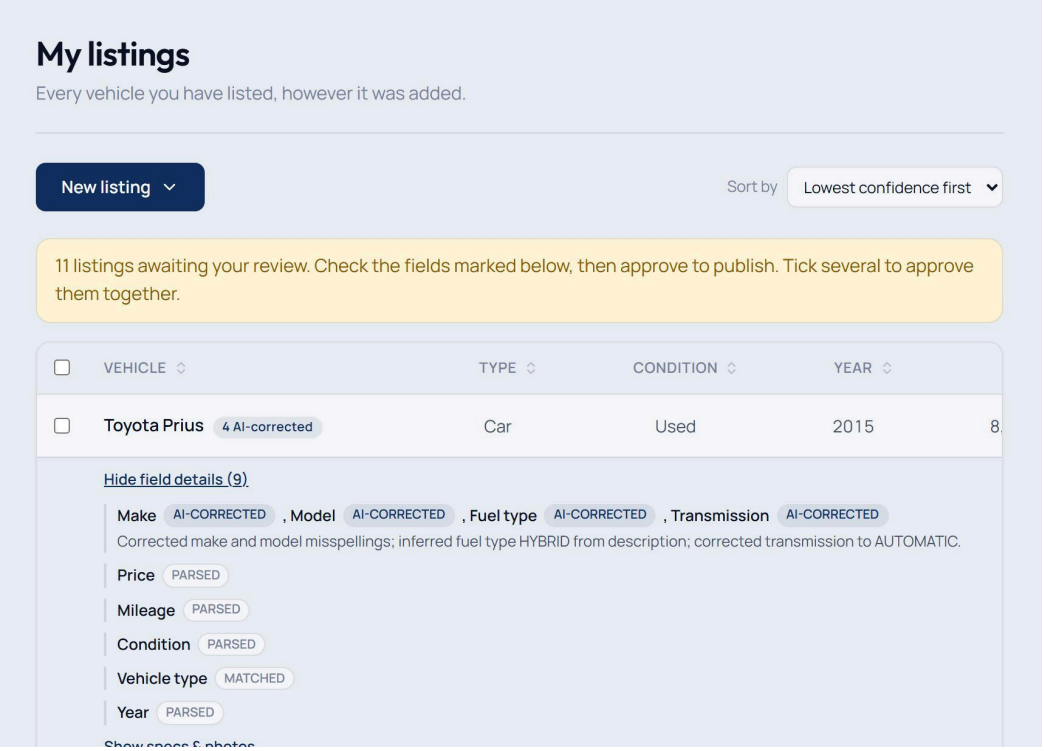


Fig. 7. My listings with the field details of an imported vehicle.

A single listing is published with **Approve**. Several listings are published together by ticking their check boxes, which displays a selection bar, and selecting **Approve selected**. After the prompt is confirmed, the platform reports how many listings were published and how many were skipped because they were no longer pending.

4.3 Editing, Archiving and Deleting

Edit opens the listing form with the current details, as shown in Fig. 8. Changes to searchable details are re-indexed automatically. Choosing new photographs replaces the existing ones, whereas a single photograph can be removed with its own button, which takes effect immediately. **Archive** removes a listing from search after a confirmation prompt, and **Unarchive** makes it live again. **Delete** is offered only for listings that were never published and asks for confirmation, because the removal of the listing and its photographs is permanent.

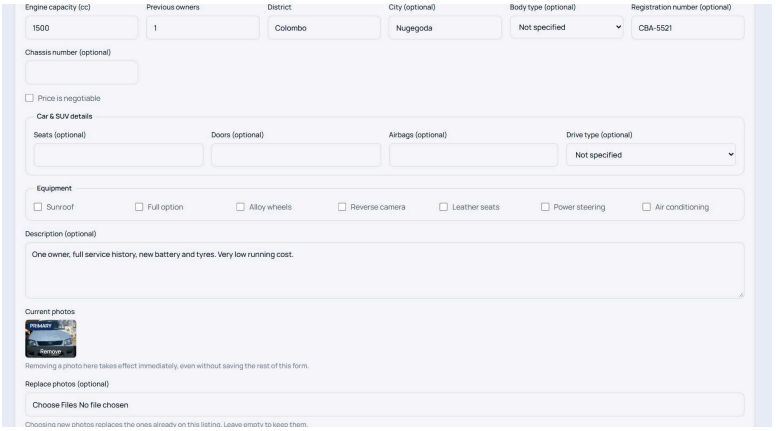


Fig. 8. Lower part of the edit form, showing the current photographs.

5. Manual Listing

5.1 Listing Form

A single vehicle is added through **Manual listing** in the sidebar or through **New listing** in **My listings**. The form is shown in Fig. 9. The vehicle type is chosen first, because it determines which further sections are displayed. The mandatory fields are the type, make, model, year of manufacture, price in LKR, mileage, fuel type, transmission, colour, engine capacity, number of previous owners and district. The registration year, condition, city, body type, registration number, chassis number and a negotiable-price option are optional. A district can be typed and selected from the suggested list of Sri Lanka's districts.

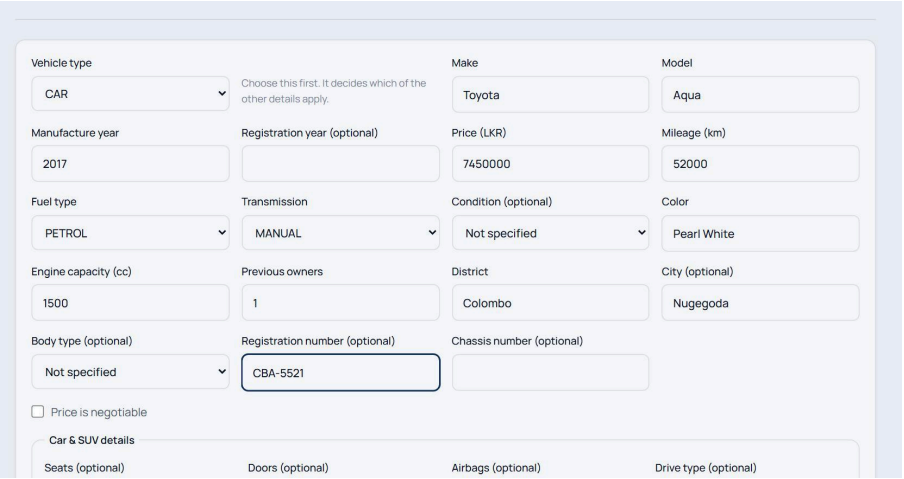


Fig. 9. Manual listing form.

5.2 Type-Specific Details, Description and Photographs

Below the common fields, a section appears for the chosen type. It collects seats, doors, airbags and drive type for cars and SUVs, stroke, cooling and start type for motorcycles, seating capacity, roof type, wheelbase and door configuration for vans and buses, and load capacity, payload, axle count and cargo bed type for trucks, lorries and pickups. An equipment section, which is hidden for motorcycles and three-wheelers, records items such as a sunroof, alloy wheels or a reverse camera. A free-text description and up to ten photographs complete the form, as shown in Fig. 10. Photographs must be in JPEG, PNG or WebP format and must not exceed 8 MB each, and the first photograph becomes the main image shown in the search results.

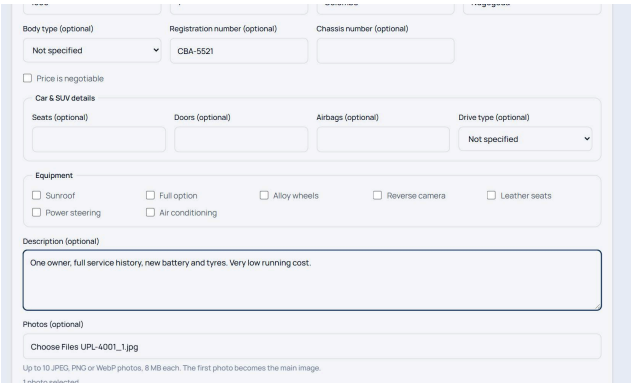


Fig. 10. Description and photograph fields of the manual listing form.

5.3 Saving

Missing or invalid fields are highlighted with a message when **Create listing** is selected, and the listing is saved only when all are valid. Uploading photographs can take a few seconds. A slow connection may delay the confirmation, and for that reason **My listings** should be checked before a listing is submitted again, so that duplicates are avoided.

6. Bulk Upload

6.1 Overview

Bulk upload imports an entire inventory in one operation and is available to business dealers only. It follows the six steps shown in Fig. 11. The dealer prepares an inventory file and, optionally, an archive of photographs, uploads them, and the platform then cleans and validates the data automatically. A result report states how many listings were created and which rows were skipped. The created listings are pending review, and the dealer finally approves them as described in Section 4.2.

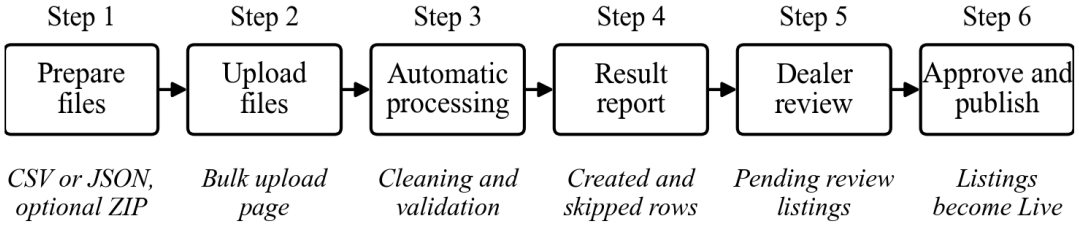


Fig. 11. Bulk upload workflow.

6.2 Preparing the Inventory File

The inventory is supplied either as a comma-separated values file, which follows the common format described in RFC 4180 [4], or as a JSON file that contains a single array of flat objects, as defined in RFC 8259 [5]. Both are encoded in UTF-8 and may not exceed 25 MB. A spreadsheet workbook in .xlsx format is not accepted and must first be saved as CSV. Each row or object describes one vehicle, and the column names are the same in both formats.

Thirteen columns are mandatory, namely make, model, year, price, mileage, fuel type, transmission, colour, engine capacity, number of owners, district, condition and vehicle type. All other columns are optional, and they include the registration number, body type, city, chassis number, description, negotiable flag and the type-specific specifications and equipment items. Values are interpreted flexibly, so that “Rs. 3,500,000”, “3.5M” and “45,000 km” are all accepted, and a misspelled make or model is corrected when it is close to a known entry. An unrecognised make or model, however, is rejected. Three buttons at the top of the upload page support the preparation. **Download template** supplies a correctly headed file, **View fields** lists every column with its accepted values, and **Known makes & models** lists the recognised makes and models so that spelling can be checked beforehand.

6.3 Preparing the Photograph Archive

Photographs are supplied as a single ZIP archive of at most 250 MB that contains JPG, PNG or WebP files. Each file is named after the registration number of its vehicle, for example CAB-1234.jpg, with a numeric suffix for further images of the same vehicle, for example CAB-1234_2.jpg. Images whose registration number does not appear in the inventory file are

skipped, and a vehicle without a registration number cannot be matched, so its photographs must be added later through the edit form.

6.4 Uploading

On the first visit, a preparation guide is displayed, as shown in Fig. 12. It can be dismissed permanently with **Don’t show this again** and reopened at any time with **Instructions**. The upload page, shown in Fig. 13, requires the file format to be chosen and the inventory file to be selected, and the photograph archive is optional. The page checks the files before sending them and rejects a file of the wrong type, an empty file or a file that exceeds the size limit, with a message that states the remedy. **Upload inventory** then sends the files, and a progress bar shows the transfer. Only one upload is processed at a time, and a dealer who opens the page while an upload is running is taken to its status page.

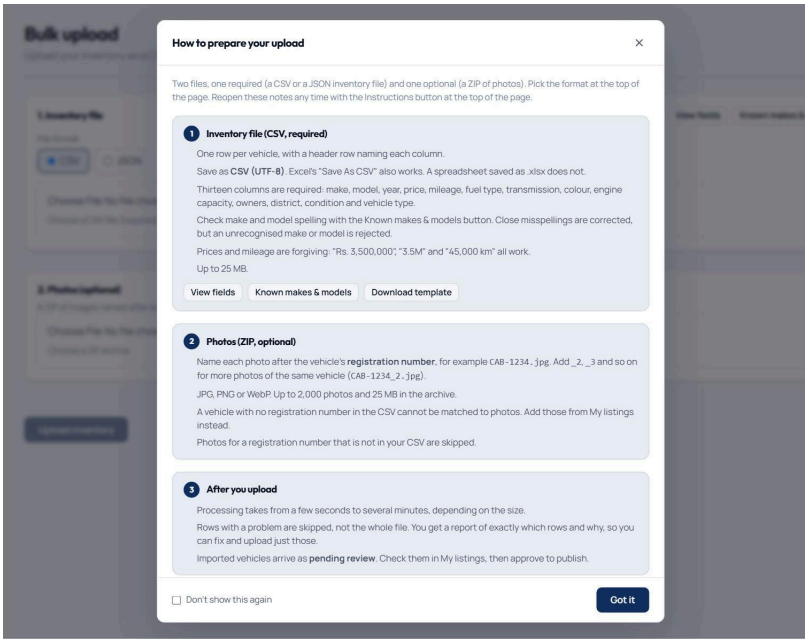


Fig. 12. Preparation guide displayed on the first visit to the bulk upload page.

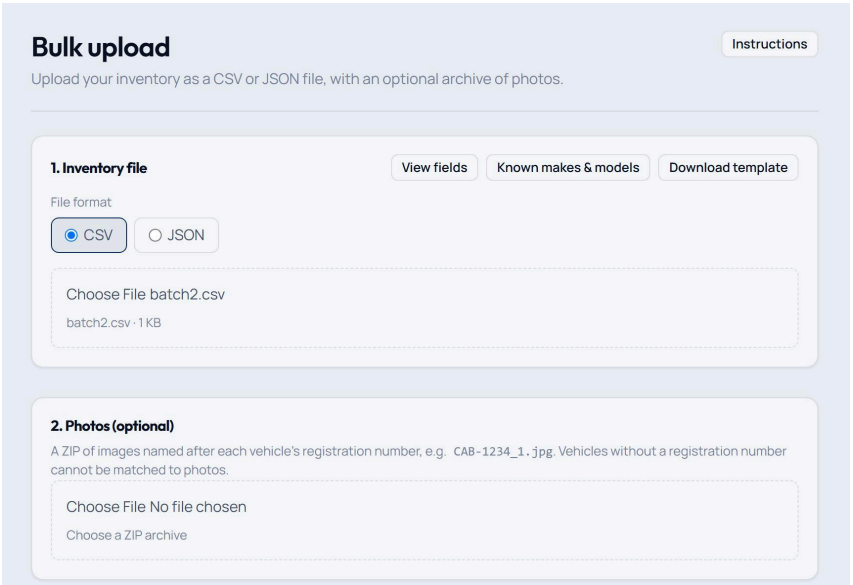


Fig. 13. Bulk upload page with an inventory file selected.

6.5 Monitoring and Results

The status page updates automatically and can be left and reopened later through **Upload history**. The possible states are summarised in Table 2. Three tiles report the rows in the file, the listings created and the rows skipped, as shown in Fig. 14. A job that ends as **Completed with skipped rows** is a successful import in which every valid row has been created. Each skipped row is listed with its row number, the reason and the data that were received, and the rows can be downloaded as a CSV file. After the **Row** and **Reason** columns have been removed and the values corrected, the file is uploaded again, and the rows that were already loaded are not affected.

Table 2. States of a bulk upload.

State	Meaning
Queued	The file has been received and is waiting to be processed.
Processing	The file is being read, matched and cleaned. A progress bar counts the rows processed.
Completed	Every row was loaded and the listings await review.
Completed with skipped rows	The valid rows were loaded and the others were skipped with reasons.
Failed	Nothing could be loaded, for example because required columns are missing or every row is invalid.

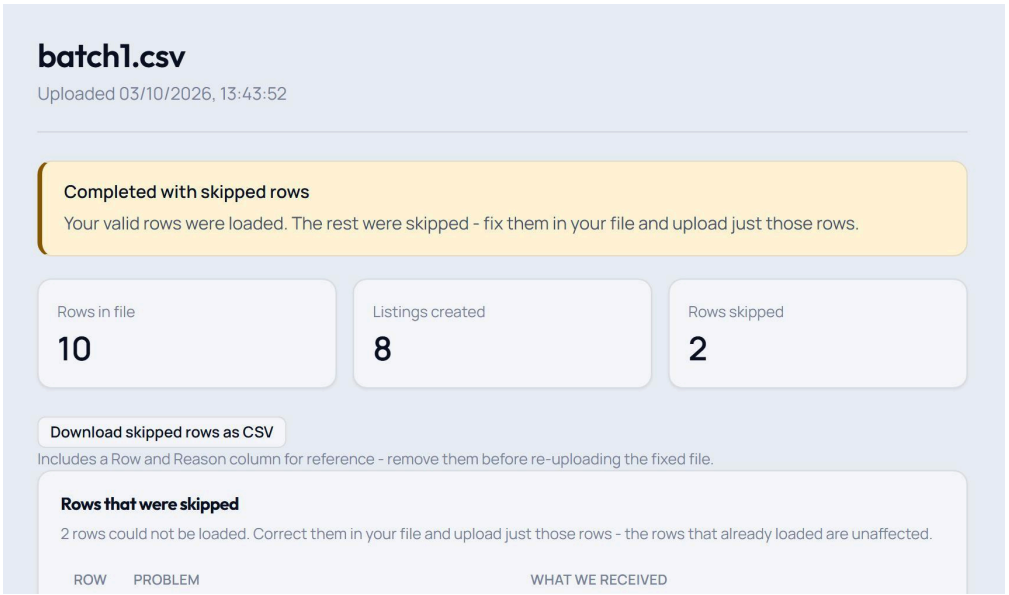


Fig. 14. Result of a bulk upload with skipped rows.

Typical reasons for a skipped row are a value outside the permitted range, such as a year before 1980, a missing mandatory value, an unrecognised make or model, and a registration number that is already listed. Fig. 15 shows a job that failed completely because every registration number in the file was already present on the platform. Each upload remains available under **Upload history**, which lists the file name, time, status and the numbers of rows loaded and skipped, with a **View details** button for the full report.

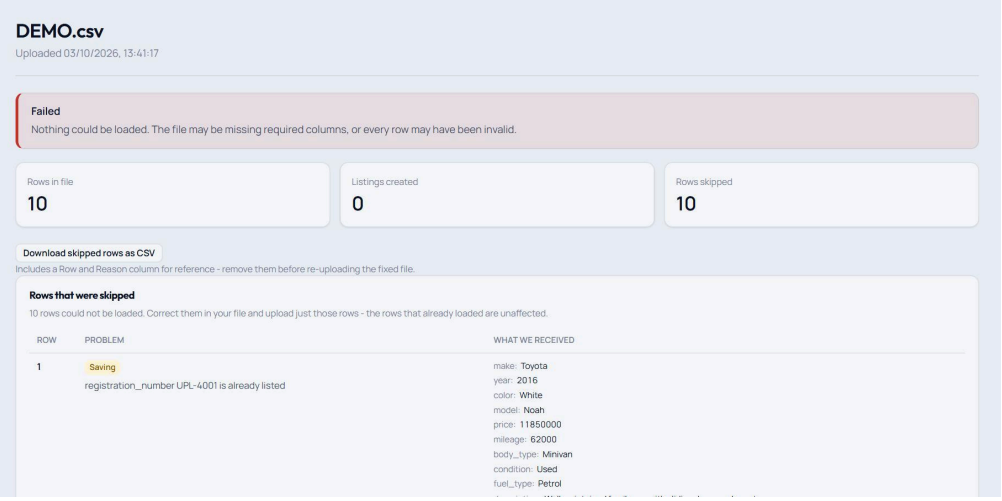


Fig. 15. Upload in which every row was rejected.

7. Troubleshooting

The most frequent difficulties and their remedies are listed in Table 3.

Table 3. Common difficulties and remedies.

Difficulty	Remedy
The portal shows no menu	The application is pending or rejected. Await the decision, or correct the details and resubmit.
Registration stops at the first step	A business must upload its certificate, and an individual must enter a valid NIC number.
Telephone number rejected	Enter the number without the leading zero, for example 771234567.
Bulk upload is not shown	Bulk upload is available to business dealers only.
Make or model rejected	Check the spelling with **Known makes & models**. A genuinely new make must be added by an administrator.
Photographs do not attach	Name each file after the registration number exactly and ensure that the number appears in the inventory file.
Imported vehicles are not in search	Imported vehicles are pending review and must be approved in **My listings**.
A live listing cannot be deleted	Only unpublished listings can be deleted. Archive the live listing instead.

References

[1] Systems and Software Engineering—Design and Development of Information for Users, ISO/IEC/IEEE Standard 26514:2022, 2022.

[2] Group 25, “Software Requirements Specification: Second-Hand Vehicle Marketplace with Intelligent Search and Automated Inventory Processing,” Dept. Comput. Sci. Eng., Univ. Moratuwa, Moratuwa, Sri Lanka, Project Rep. PID 11, Ver. 1.0, 2026.

[3] Group 25, “Software Architecture Document: Second-Hand Vehicle Marketplace with Intelligent Search and Automated Inventory Processing,” Dept. Comput. Sci. Eng., Univ. Moratuwa, Moratuwa, Sri Lanka, Project Rep. PID 11, Ver. 2.0, Aug. 10, 2026.

[4] Y. Shafranovich, “Common format and MIME type for comma-separated values (CSV) files,” Internet Eng. Task Force, RFC 4180, Oct. 2005.

[5] T. Bray, Ed., “The JavaScript Object Notation (JSON) data interchange format,” Internet Eng. Task Force, RFC 8259, Dec. 2017.